

Phylogenetics

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Phylogeny

- Greek '*phylo*' meaning tribe/clan, and '*genetikos*' meaning genesis/origin/birth
- Widely used bioinformatics application to organize our knowledge of biological diversity we observe on earth.
- Diagram that depicts lines of evolutionary descent of different species.



Why do we care
about
phylogenetics?



Case studies

SARS
(2002)

MERS
(2012)

Darwin's
finches
(2019)

Covid-19
(2019)

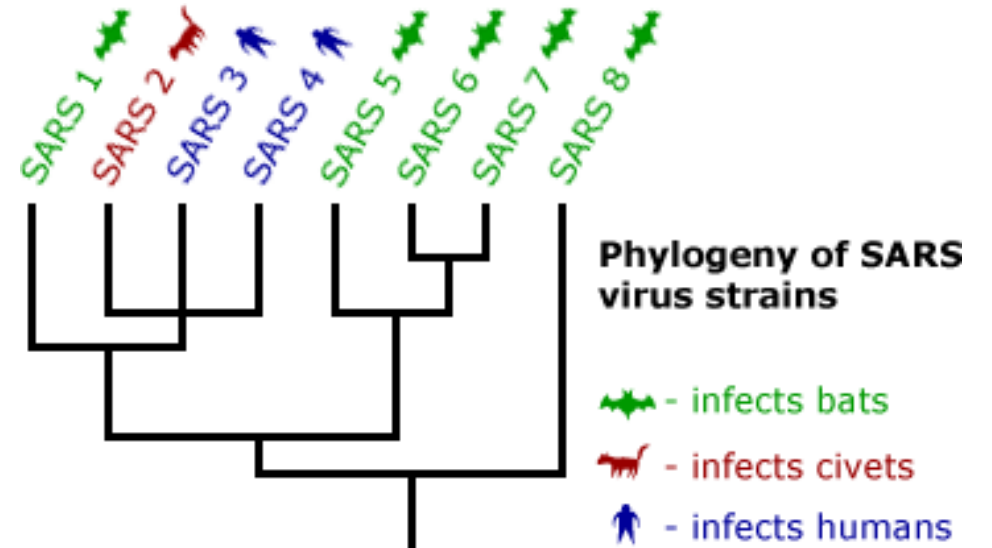
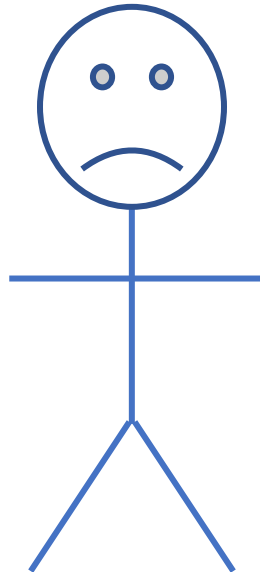


SARS (2002 - 2003)

Severe Acute Respiratory Syndrome

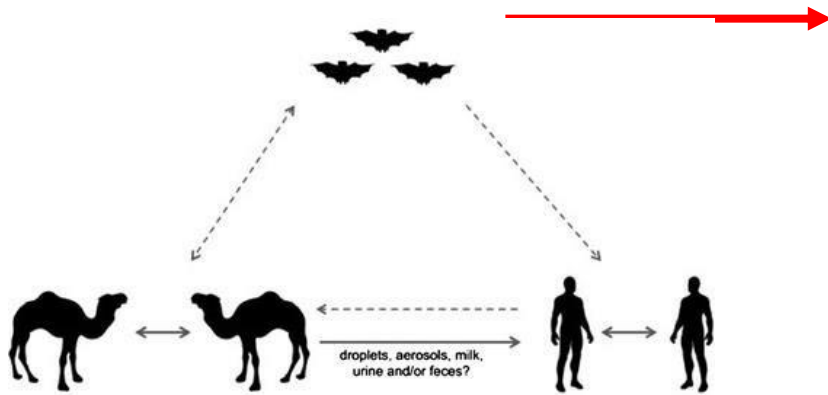


Chinese horseshoe bat

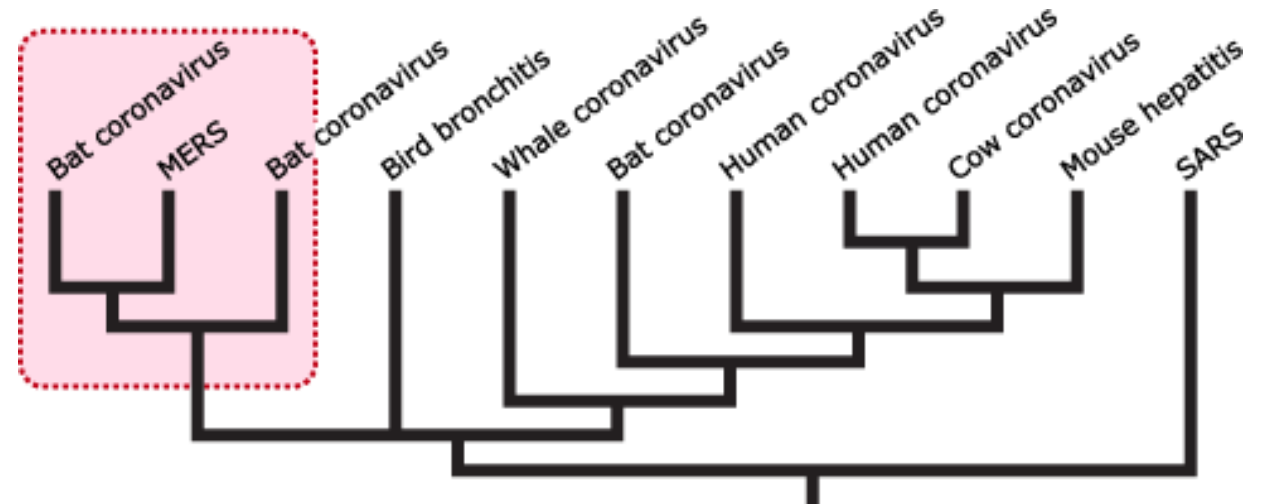


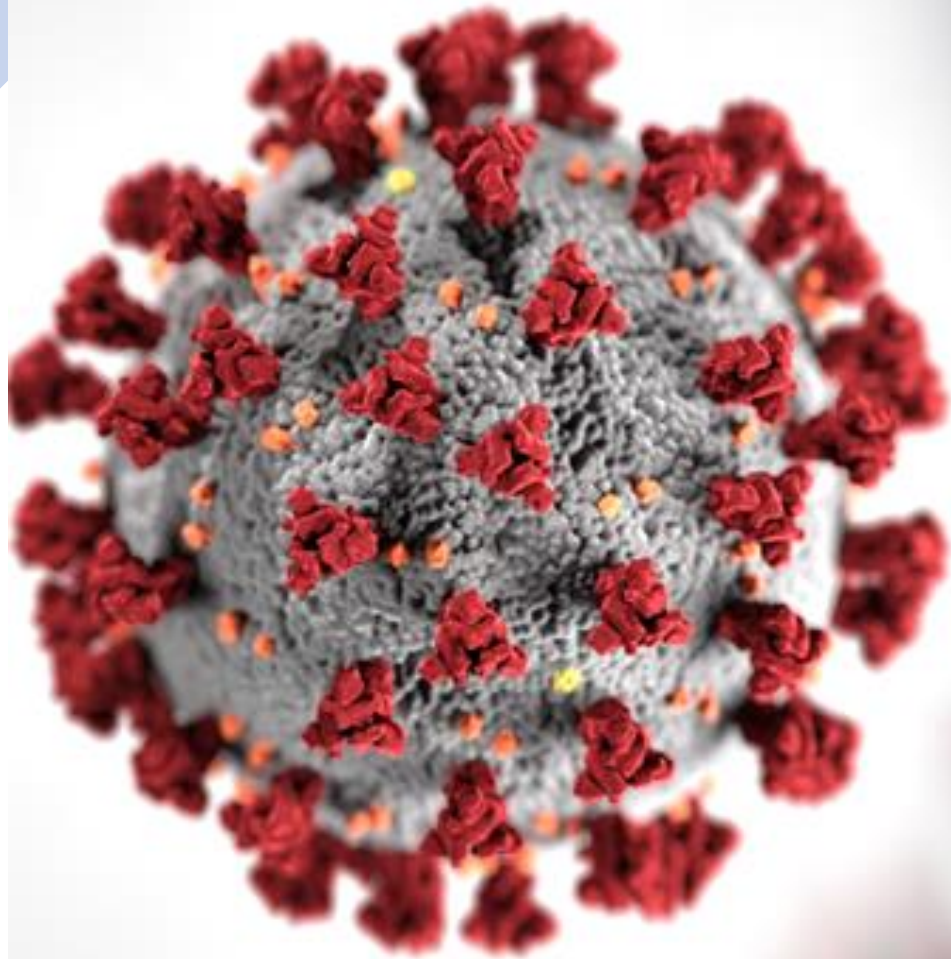
MERS (2012 - 2013)

Middle East Respiratory Syndrome



Solid line, documented direction of MERS-CoV transmission
Dashed line, hypothetical direction of MERS-CoV transmission



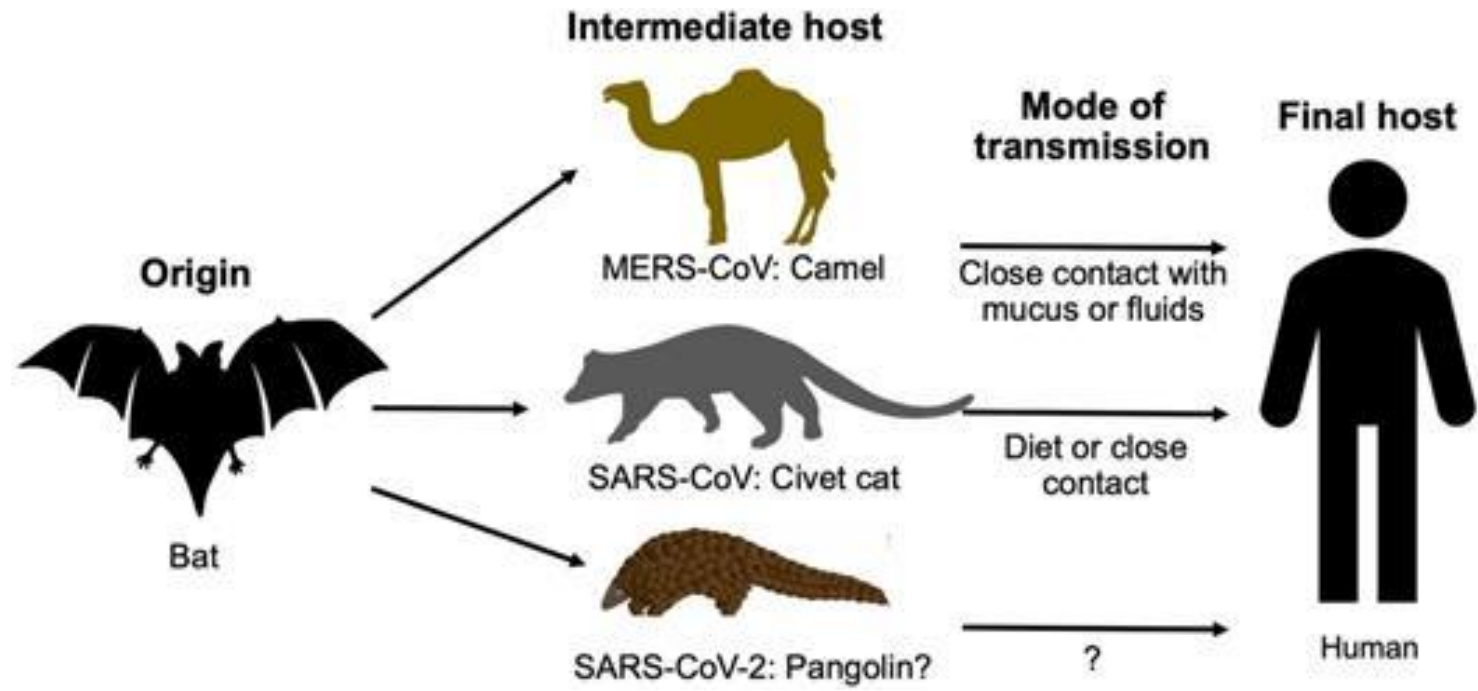


COVID-19

CORONAVIRUS DISEASE 2019

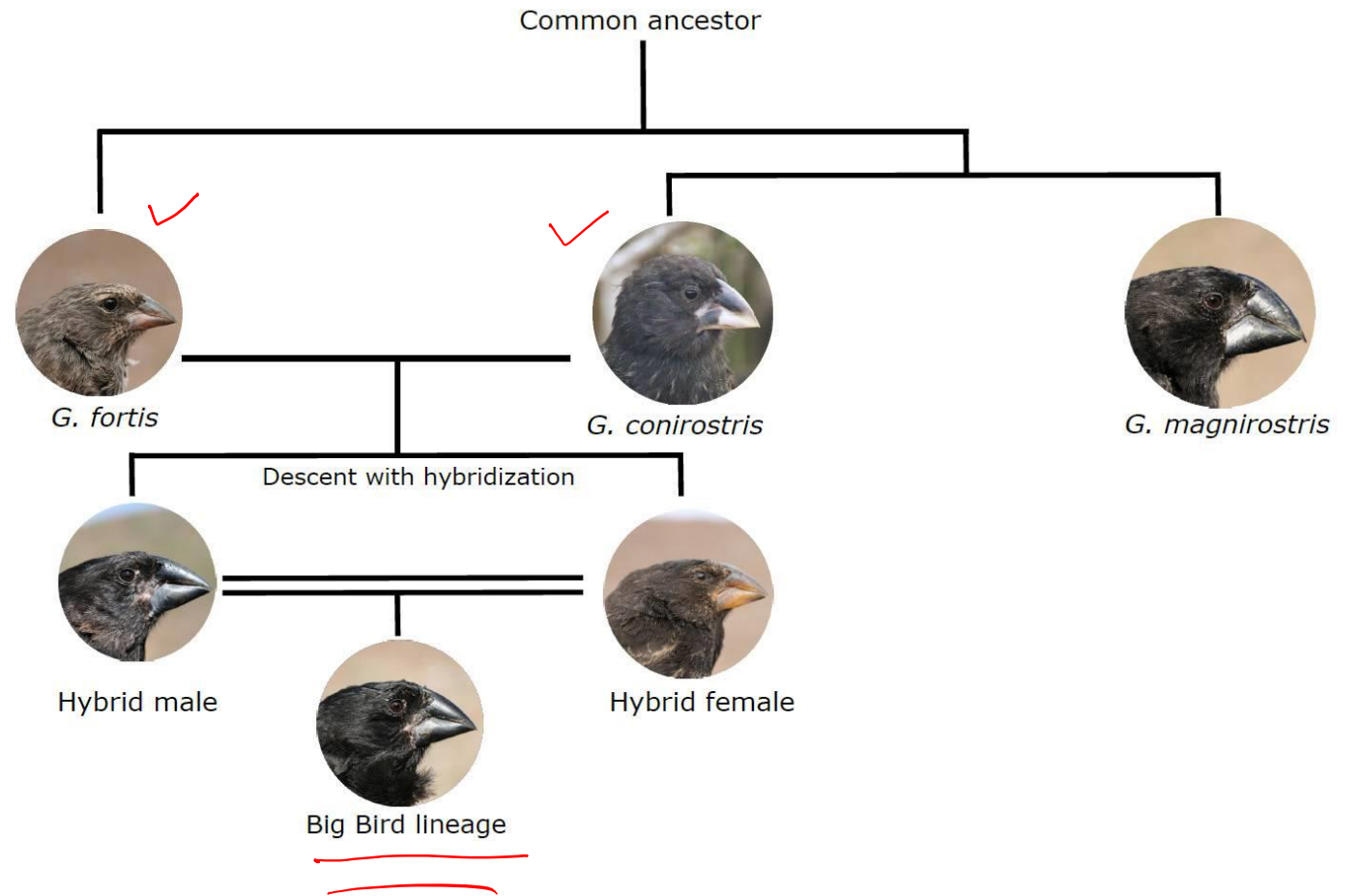


Why bats?



Origin of the *big bird lineage*

(1981 - 2019)



Reference: <https://www.princeton.edu/news/2017/11/27/study-darwins-finches-reveals-new-species-can-develop-little-two-generations>

